THE PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA REGION

FORM ~~TWO~~ TWO REGIONAL EXAMINATION

CODE: 034 AGRICULTURE

MARKING SCHEME

SECTION A

1. MULTIPLE CHOICE (10 Marks)

Qns	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answ	A	D	B	C	B	C	C	C	C	A

2. MATCHING ITEMS (5 Marks)

Qns	i	ii	iii	iv	v
Answ	E	B	C	D	A

SECTION B

3(a) What do you understand the following terms as used in agriculture:

(i) Soil → Is the thin upper layer of the earth's crust.

(ii) Agriculture → Is the art and science of raising or growing crops and keeping livestocks.

(iii) Agrotourism → Is the part of agriculture that allows people to learn and enjoy farming activities. People can visit farmers and learn about farming for leisure.

(a) (iv) Soil profile → is the vertical section through which all horizons are extended from the top soil to the parent materials or bed rock. (5 MARKS)

(v) Soil texture → is the courseness and fineness of the soil particles.

3(b) (1) Biology – In biology we learn about living ~~things~~ things such as livestocks Crop plants and Animals. This knowledge helps to understand plant and animal life, pests, parasites and diseases with their control and genetics for better breeds

(2) Chemistry – The knowledges and skills from chemistry is essential for soil testing, manufacturing and application of fertilizers and agrochemicals for agriculture production

(3) Mathematics → Mathematical operation such as addition, substraction, multiplication and division are used in agriculture for calculating plant distances, livestock ratios and farm finances. (5 Marks)

4(c) Five importances of horticulture

(i) provides nutritional food
(ii) Source of local and foreign incomes
(iii) Employment creation
(iv) Stimulate industrial grow
(v) Soil conservation
(vi) Provide business opportunities

(5 Marks)

(a) Identify critical stages during which the crops demand significant water.

(i) Germination/establishment stage

(ii) Flowering stage

(iii) Grain, seed, tuber or fruit formation stage (3 marks)

~~(iv)~~

4 (b) How does Plant Cover crops helps to retain moisture in the soil?

⇒ Application of dry grass, banana leaves or crop residues help to conserve moisture and suppresses weed by covering the soil ~~from direct heating~~ and preventing it from direct heating by the sun, Also reduces evaporation of soil moisture by maintain low temperature over the soil. (2 Marks)

5(a) Poultry species: Should be exotic breed

(b) Suitable system of production: Should be intensive System

(c) Reasons:
- less feed wastage and space
- low diseases spreads.
- Easier egg collection and management
- Vice like egg eating can be easy identified (10 Marks)

6 Post-harvesting handling includes all activities done after harvesting, Briefly ~~Explain~~ explain how does these activities is done.

(i) Cleaning: Wash the crops produces soon after being harvested to remove debris and dirtiness

(ii) Drying: Reducing moisture content to prevent spoilage

(iii) Sorting and grading: Separating crops based on quality

(iv) Storage: Keeping crops under suitable condition

(v) Processing: Converting crops into other products

(vi) Packaging: Proper packing for storage and transportation

(vii) Transporting: Movements of the produce/ products from the field to the storage structure or market.

(viii) Marketing: ~~Food~~ Products can be marketed at farm gate prices on the field, in the local market and international ma (10M

7(a) List five(5) factors to consider when selecting good site for crop production.

(i) Fertile soil with good drainage
(ii) Suitable climate
(iii) Accessibility
(iv) Availability of water for irrigation
(v) low incidence of pests and diseases
(vi) Security of the area. (5 Marks)

7(b) List down at least four characteristics of good planting Materials.

(i) Free from pests and diseases
(ii) High germination capacity
(iii) High yield potential
(iv) Suitable for the local environment. (5 Marks)

8(a)

TYPE OF CROP	PLANTING MATERIALS
(i) Cassava	Stem cutting
(ii) Banana	Suckers
(iii) Round potato	Stem tubers
(iv) Tomato	seeds
(v) Sweet potato	Cuttings/Vines

(5 Marks)

8(b) Tools and implements used in land clearing and preparation

(i) Slashers (ii) Hoes ~~(iii) Mat~~
(iii) Machetes (iv) Axes (v) Fork
(vi) Ox-ploughing (vii) Tractor-mounted plough
(viii) Harrows (ix) Ridger sets. (5 Marks)

9(a) Data given

Spacing = 4M x 3M = $12M^2$ per plant area

1 hectare = $10,000M^2$ and area of the plant is $12M^2$

∴ Plant population = $\frac{\text{Field area}}{\text{Plant area (spacing)}}$

∴ Plant population = 833.3 ≈ 834 plant/ha (5 Marks)

9(b)

PLANT PART HARVESTED	EXAMPLE OF THE CROPS
1. Roots	Cassava, Carrot, sweet potato.
2. Tubers	Round potatoes
3. Stems	Sugarcane
4. Leaves	Chinese cabbage, kale, Amaranths
5. Seeds	Maize, Beans.

(5 Marks)

Qn 10

1.1 Branches of agriculture

(Clear introduction ~~3 marks~~ (3 marks))

(i) Crop production

(ii) ~~Crop~~ Animal production

(iii) Soil science

(iv) Agricultural engineering

(v) Agricultural economics

(10 Marks with 2 Mark @ Points)

(Clear conclution = 2 Marks)